Mohammed Fawwaz Ali

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.08
Narayana Junior College	Class XII (TSBIE)	2023	98.30%
Vidyanjali High School	Class X (CBSE)	2021	95.40%

WORK EXPERIENCE

Goldman Sachs - Summer Analyst SDE

- Engineered backend services for an **SRE monitoring dashboard** using **Java** and **Spring Boot**, enabling operational telemetry and status tracking of jobs and infrastructure metrics.
- Designed and implemented **REST APIs** to expose database metrics, file-server information, and job status updates.
- Integrated **near-real-time monitoring** across 3 database systems, tracking key metrics per instance with **threshold-based checks** to flag potential reliability issues.
- Developed a **filesystem-based workflow tracking mechanism**, correlating state files across **5-10 major states** and **15-20 sub-states** to reconstruct job progression and surface accurate status updates on the dashboard.
- Contributed to the **React frontend** by integrating backend APIs and supporting UI updates for file-server metrics and workflow data.
- Debugged and resolved build and deployment issues in the development environment using **GitLab CI/CD pipelines** to ensure code quality and maintainability.

Technologies: Java, Spring Boot, REST APIs, React, GitLab, Postman

SKILLS

- **Languages & Databses:** C, C++, Java, Python, HTML, CSS, Javascript, SQL, MySQL, MongoDB
- **Libraries & Framewords:** C++ STL, Spring Boot, Node.js, Express.js, React, FastAPI, NumPy, pandas, scikit-learn
- **CoreConcepts:** Data Structures, Algorithms, OOP, DBMS, OS, LLD

PROJECTS

PR Assist - GitHub

Python, FastAPI, React, PostgreSQL, pgvector, RAG, Semgrep, Docker

- Built an AI-powered **Pull Request review assistant** that complements **CI/CD** by analyzing code changes and surfacing potential bugs, security vulnerabilities, and business-logic issues before human approval.
- Engineered an **AST-aware RAG pipeline** that parses repositories into symbol-level chunks and retrieves relevant repository context for each Pull Request, enabling the LLM to analyze changes beyond the modified code.
- Implemented a **verification layer**, deterministic checks first, LLM escalation only for ambiguous cases, that **validates every AI-generated** finding against source code and static-analysis results before surfacing the final review.
- Evaluated the review pipeline across **20 PRs**, achieving **75% Recall@5** for repository-context retrieval and **80% PR-level verdict accuracy**, with an **average end-to-end review latency of 40s.**

Stavio - GitHub

Javascript, Raect, Node.js, Express.js, MongoDB, Redis, k6

- Developed a **full-stack property listing platform** with a decoupled frontend-backend architecture and RESTful APIs, implementing session-based authentication, role-based authorization, server-side validation, and structured data modeling.
- Implemented **concurrency-safe and idempotent booking workflows** using MongoDB transactions and atomic availability checks, preventing duplicate bookings from simultaneous requests and request retries.
- Implemented **Redis-backed response caching** and load tested the deployed API with **k6**, increasing throughput from **8.8 to 16.4 req/s** and reducing **P95 latency by ~53%** with 0% request failures.

ML Pipeline - GitHub

Python, scikit-learn, pandas, NumPy, PyYAML

- Engineered a **modular, configurable ML pipeline** for tabular data, integrating **data ingestion, schema validation, preprocessing, model training, and evaluation** into reusable end-to-end components.
- Implemented **automated data-quality validation** for schema consistency, missing values, duplicates, data types, and feature ranges, with reusable preprocessing and train-only fitting to prevent data leakage.
- Built a **cross-validated model comparison framework** with configurable evaluation metrics, automatically **selecting and persisting the best-performing model** along with preprocessing artifacts and model metadata.

SCHOLASTIC ACHIEVEMENTS

- Selected for **Google** India's Student Upskilling Launchpad and successfully completed the program.

RELEVANT COURSES

- **Computer Science**: Introduction to Programming, Introduction to Modern AI, Business Analytics using Python
- **Mathematics:** Probability, Statistics, Linear Algebra, Differential Equations

POSITIONS OF RESPONSIBILITY

Placement, Internship Coordinator, Office of Career Services

Programming & ML Core, Robotix (theRoboticsClub)

Sponsorship & Finance Head, MAE Department Day

Events Manager, Entrepreneurship Cell

Techy Deputy Coordinator, Elan and nVision (the annual Techno-Cultural Fest)